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MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER Discipline Specific Elective (DSE) - MG4DSECHE200 - POLYMER CHEMISTRY

1. Define the chain initiation step in free radical polymerisation and provide a chemical equation for it.

The initiation of the free radical polymerization occurs in two steps

A, the decomposition of initiator to produce the free radical

B, addition of the free radical to the monomer with the generation of another radical

I → R**.** +R**.**

R. + CH2=CHX -> R-CH2X-CHX**.**

R-CHX-CH2**.**

2. What are shape-memory polymers?

Shape memory polymers are a class of smart materials that can remember and recover their Original shape after being deformed into a temporary shape when exposed to an external stimulus such as heat, light ,PH, electric or magnetic field, moisture etc

Examples : PU,PCL,EVA

3. Give an example for a monomer which undergoes cationic polymerization explain why it undergoes this type of polymerization.

Styrene have electron releasing groups which undergo cationic polymerisation due to stable carbocation intermediate as well as due to double bond which also donate electrons.

Examples: isobutylene, styrene, methyl styrene

4. What is atactic polymer? Give an example.

if the R groups on successive stereocentres are randomly distributed on the two sides of the planar zig zag polymer chain, the polymer does not have an order and is termed atactic

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An atactic polymer is a type of polymer in which the substituent groups (side groups attached to the main chain) are arranged randomly along the polymer backbone. There is no regular stereochemical order (unlike isotactic or syndiotactic polymers).

Examples: atactic polypropylene

5. Give two examples of polymers prepared by suspension polymerisation.

Polyvinyl chloride (PVC), Poly styrene

styrene acrylonitrile , polyvinyl acetate beads

6. Explain the effect of plasticizers on glass transition temperature.

Plasticizers reduce glass transition temperature due to reduction in cohesive force of attraction between polymer chains. Attractive forces reduce the cohesive forces between the polymer chains resulting in increased flexibility and softness, reduced brittleness and improved processability. Plasticizers make polymer chains move more freely, so the material becomes flexible at a lower temperature, hence T_g is reduced.

7. What structural change occurs during vulcanisation of rubber?

Vulcanisation is a chemical process in which long chains of rubber molecules become crosslinked by reactions with the vulcanizing agent to form 3D structures. The reaction transforms the soft, weak, plastic material into strong elastic product. The rubber loses its tackiness and becomes insoluble in solvents and more resistant to deterioration.

8. Identify the polymers from the monomers: (i) caprolactum and (ii) hexamethylene diamine + adipic acid

Caprolactum - nylon 6

Hexamethylene diamine and adipic acid- Nylon -6,6

9. Identify the monomers used to make PET polymers.

ethylene glycol+ terephthalic acid

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10. Describe the term polymer reagent with reference to polymer support.

A polymer reagent is a polymer containing a functional group that acts as a reagent to bring about a chemical transformation as low molecular weight materials. They contain functional group that performs a particular chemical function.

11. How can temperature sensitive polymers be utilized to achieve targeted drug delivery?

Temperature-sensitive polymers can release drugs when they reach a specific temperature such as body temperature. This property helps in targeted and controlled drug delivery systems. Tumor cells have slightly high temp than normal cells.

12. Compare the electrical conductivity of polyacetylene with other conducting polymers.

Polyacetylene shows high electrical conductivity when it is doped with suitable agents. Its conductivity is higher compared to many other conducting polymers due to its conjugated structure.

PART B

13. Comment on the importance of DNA as a polymer.

DNA is a natural polymer composed of repeating nucleotide units. DNA is a **biopolymer** made up of repeating units called **nucleotides**. Each nucleotide contains:

- A sugar (deoxyribose)
- A phosphate group
- A nitrogenous base

These nucleotides are linked to form a long chain, giving DNA the characteristics of a high molecular weight polymer. It plays a crucial role in storing and transmitting genetic information in living organisms. The structure of DNA consists of a double helix formed by complementary base pairing. It controls protein synthesis through transcription and translation processes. DNA has the unique ability to replicate itself accurately during cell division. It is essential for heredity and variation in organisms. DNA is widely

used in genetic engineering and biotechnology applications. It also has applications in forensic science for identification purposes.

14. Explain the hydrolytic ring-opening polymerisation mechanism of Nylon-6.

Nylon-6 is prepared by the hydrolytic ring-opening polymerization of caprolactam. In the first step, caprolactam reacts with water to form aminocaproic acid. This step involves the opening of the cyclic amide ring structure. The formed amino acid then undergoes condensation polymerization. Repeated condensation reactions lead to the formation of long polymer chains. The process results in the formation of amide linkages between monomer units. Water is eliminated during the condensation step.

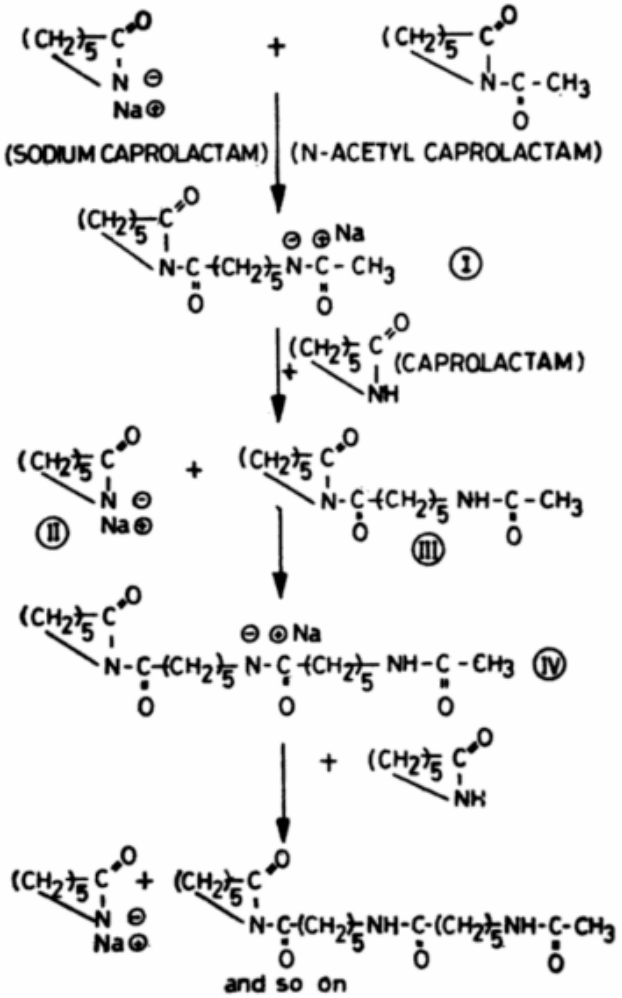


Fig. 2.2. Ring-opening polymerisation of caprolactam.

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15. Compare solution polymerisation and emulsion polymerisation.

Solution polymerization is carried out in a suitable solvent where both monomer and initiator are soluble. Emulsion polymerization is carried out in water using surfactants to form micelles. In solution polymerization, the polymer remains dissolved in the solvent. In emulsion polymerization, the polymer is obtained as a latex. The rate of polymerization is generally slower in solution polymerization. Emulsion polymerization has a faster reaction rate due to high surface area. Heat control is difficult in solution polymerization. emulsion polymerization provides better heat control.

16. Explain the factors influencing the crytallinility and amorphous characters of polymers?

Linear polymers tend to be more crystalline than branched polymers. Symmetrical structures promote better packing and higher crystallinity. Strong intermolecular forces also enhance crystallinity. Branching in polymer chains reduces the ability of chains to pack closely. The rate of cooling during solidification also affects crystallinity. Slow cooling allows better arrangement of polymer chains. Molecular weight also influences crystallinity to some extent.

17. Describe the structure and properties of Urea–formaldehyde resin

Urea–formaldehyde resin is formed by the condensation of urea and formaldehyde. It contains methylene linkages connecting nitrogen atoms. The polymer forms a highly cross-linked network structure. It is hard, rigid, and brittle in nature. The resin has good adhesive properties. It also shows resistance to heat and chemicals. It is widely used in plywood and laminates. It is used as an adhesive in furniture industries. it is a thermosetting polymer.

PROPERTIES:

1. MECHANICAL PROPERTIES

- Hard, rigid, and strong
- Good surface finish
- Brittle in nature

2. THERMAL PROPERTIES

- Good heat resistance

- Does not soften on heating (thermosetting behavior)

3. ELECTRICAL PROPERTIES

- Excellent **electrical insulator**
- Widely used in electrical fittings

4. CHEMICAL PROPERTIES

- Resistant to oils, solvents, and weak acids
- Poor resistance to strong acids and alkalis

5. OTHER CHARACTERISTICS

- Light-colored or transparent
- Can be easily molded before curing
- Low cost and widely available

18. Give the structure of polystyrene and discuss a few of its applications.

Polystyrene is a polymer formed from condensation of styrene monomers. Its structure consists of a hydrocarbon backbone with phenyl groups attached.

Its repeating unit is:

$[-CH_2-CH(C_6H_5)-]n$

- The backbone consists of a carbon–carbon chain.
- Each repeating unit has a phenyl group (–C H) attached to every second carbon atom.

The presence of phenyl groups makes the polymer rigid. It is lightweight and has good insulating properties. Polystyrene is transparent and brittle in nature. It is widely used in packaging materials. It is also used to make disposable cups and containers. It finds application in insulation and electronic goods.

19. Describe the application of Nomex in the field of protective clothing.

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Nomex is an aromatic polyamide known for its high thermal stability. It does not melt easily and shows excellent flame resistance. The polymer maintains its strength even at high temperatures. It is lightweight and comfortable for wearing. Nomex is widely used in firefighter suits. It is also used in military and industrial protective clothing. The material provides protection against heat and fire hazards. It also has good chemical resistance.

20. Discuss the applications of poly vinyl carbazole (PVK) in photoconducting polymers.

Polyvinyl carbazole is a photoconducting polymer. It has the ability to conduct electricity upon exposure to light. This property makes it useful in xerographic processes. It is used in photocopiers and laser printers. PVK is also used in organic electronic devices. It has good charge transport properties. It can be used in photovoltaic cells. The polymer is stable and easy to process.

PART C

21. Explain in detail the classification of polymers based on origin and structure.

Polymers can be classified based on their origin and structure. Based on origin, polymers are divided into natural, semi-synthetic, and synthetic polymers. Natural polymers occur in nature and include materials like cellulose and proteins. Semi-synthetic polymers are chemically modified natural polymers such as cellulose acetate. Synthetic polymers are man-made and include plastics like polyethylene and PVC.

Based on structure, polymers are classified as linear, branched, and cross-linked polymers. Linear polymers are the polymers where monomeric units are linked together to form long straight chains. The polymeric chains are stacked over one another to give a well packed structure. As a result of close packing, such polymers have high densities, high tensile strength and high melting points. Common examples of these type polymers are polyethylene, polyester and nylon etc.

Branched Chain Polymers: In this type of polymers, the monomeric units are linked to constitute long chains, which are also called main-chain. There are side chains of different lengths which constitute branches. Branched chain polymers are irregularly packed and thus, they have low density, lower tensile strength and lower melting points as compared to linear polymers. Amylopectin and glycogen are common examples of such type.

Cross-Linked Polymers or Network Polymers: In this type of polymers, the monomeric units are linked together to constitute a three dimensional network. The links involved are called cross links. Cross-linked polymers are hard, rigid and brittle because of their network structure. Common examples of this type of polymers are bakelite, formaldehyde resin, melamine, etc

22. Explain in detail the practical significance of polymer molecular weight?

Molecular weight is an important parameter that determines the properties of polymers. It affects the mechanical strength of the material significantly. Polymers with high molecular weight generally have higher strength and toughness. It also influences the viscosity of polymer solutions and melts. Higher molecular weight leads to increased viscosity. The melting point and glass transition temperature are also affected by molecular weight. As molecular weight increases, thermal stability improves. Molecular weight also affects the processability of polymers. High molecular weight polymers are difficult to process due to their high viscosity. Low molecular weight polymers are easier to process but have lower strength. Molecular weight is measured using number average and weight average methods. It is important in industrial applications for quality control. It helps in selecting suitable polymers for specific uses.

23. Describe the properties and applications of PMMA.

Polymethyl methacrylate is a transparent thermoplastic polymer. It is commonly known as acrylic or organic glass. PMMA has excellent optical clarity and allows transmission of light. It is lightweight compared to glass and has good impact resistance. The polymer shows good weather and UV resistance. However, it is brittle and can crack under stress. PMMA has good electrical insulating properties. It is easy to mold and shape into different forms. It is widely used as a substitute for glass. Applications include lenses, windows, and display panels. It is used in aircraft windows due to its light weight. It is also used in signboards and decorative items. PMMA is used in medical devices and optical instruments.

24. Discuss the structure-property relationship, and applications of biodegradable polyurethanes.

Biodegradable polyurethanes are polymers containing urethane linkages. They are designed to degrade under biological conditions. Their structure consists of soft and hard segments. Soft segments provide flexibility while hard segments provide strength. The presence of ester bonds enhances biodegradability.

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These polymers are biocompatible and safe for medical use. They degrade into non-toxic products in the body. The rate of degradation can be controlled by modifying the structure. They show good mechanical properties and elasticity. Biodegradable polyurethanes are used in medical implants. They are also used in drug delivery systems. They find applications in tissue engineering and wound healing. These polymers are used in surgical sutures. It is made from condensation reaction between diol and diisocyanate